FreqForcing: Autoregressive Long Video Generation via Spectral Self-Anchoring
Supplementary Material

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Overview

We provide more details and results in this supplementary material, including the following:

- Details of Short-Time Fourier Transform (STFT) analysis for self-attention outputs and decoded pixel frames.
- Visualization of distribution shift for comparison.
- Additional ablation studies.
- More visual results on *120s* videos.
- Limitations and future work.

Details of Short-Time Fourier Transform

We analyze the relative spectral drift in both the self-attention output and decoded pixel frames using a sliding-window Short-Time Fourier transform (STFT). Unless otherwise specified, we use a Hann window of $W = 16$ samples, a hop size of $H = 4$, and no zero-padding. All curves are averaged over 36 prompts from MovieGen (Polyak et al. 2024) and normalized by the corresponding energy in the first window.

Self-Attention Outputs. For each latent chunk, we collect the self-attention output at denoising step t_4, before the attention output projection. For methods without spectral anchoring, we use the local attention output. Let $\mathbf{A}_{f,p,h,d}$ denote the attention output at latent frame f, spatial token p, attention head h, and head channel d. We first compute the spatial DC component of each frame:

$$\mathbf{a}_{f,h,d} = \frac{1}{N_s} \sum_{p=1}^{N_s} \mathbf{A}_{f,p,h,d}. \tag{1}$$

The head and channel dimensions are flattened to obtain $\mathbf{a}_f$. For Fig. 3 in the main paper, we use the output of the 29th DiT layer. To suppress the rollout-wide constant offset when analyzing non-zero temporal frequencies, we subtract the sequence-wide feature mean:

$$\widetilde{\mathbf{a}}_f = \mathbf{a}_f - \frac{1}{F} \sum_{j=1}^{F} \mathbf{a}_j. \tag{2}$$

For the m-th sliding window, the STFT is computed independently for every feature channel:

$$Z_{m,k,c} = \sum_{n=0}^{W-1} w_n \widetilde{\mathbf{a}}_{mH+n,c} \exp\left(-\mathrm{i}\frac{2\pi kn}{W}\right), \tag{3}$$

where w_n is a symmetric Hann window. The power at each temporal-frequency bin is averaged over the $N_h \cdot D$ feature channels:

$$P_{m,k} = \frac{1}{N_h \cdot D} \sum_{c=1}^{N_h \cdot D} \left\|Z_{m,k,c}\right\|^2. \tag{4}$$

With $W = 16$, the real-valued FFT produces nine bins $k = \{0, \ldots, 8\}$. We exclude $k = 0$ from the non-zero frequency bands and define the low-frequency energy as $E_m^{\mathrm{low}} = \sum_{k=1}^{2} P_{m,k}$, and high-frequency energy as $E_m^{\mathrm{high}} = \sum_{k=3}^{8} P_{m,k}$. Moreover, the DC curve is calculated as follows:

$$E_m^{\mathrm{DC}} = \left\| \frac{1}{W} \sum_{n=0}^{W-1} \mathbf{a}_{mH+n} \right\|_2^2. \tag{5}$$

Pixel-Space Frames. After autoregressive generation, the latent sequence is decoded by the VAE into RGB frames $\mathbf{I}_t \in [0,1]^{3\times H_p \times W_p}$. To retain coarse spatial color and appearance information while keeping a compact descriptor, each RGB frame is adaptively average-pooled to an 8×8 grid:

$$\mathbf{p}_t = \mathrm{vec}\left(\mathrm{Pooling}_{8\times 8}(\mathbf{I}_t)\right) \in \mathbb{R}^{192}. \tag{6}$$

This descriptor contains $3 \times 8 \times 8 = 192$ features. It is therefore a coarse spatial RGB descriptor rather than a strict whole-frame RGB vector. We then follow the above configuration to analyze the frequency-domain energy of the frames in pixel space.

Visualization of Distribution Shift

To further illustrate the deviation of videos generated by FreqForcing from those produced within the pretrained temporal horizon, we adopt a more intuitive visualization of distribution shift. Unlike previous methods that quantify quality drift by comparing the ViCLIP scores (Wang et al. 2024) of the first and last five seconds, we directly measure the discrepancy between frames generated near the end of a

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long video and those generated within the pretrained horizon. Specifically, we perform principal component analysis (PCA) (Shlens 2014) on the frame features within the pretrained horizon and retain the first two principal components, PC1 and PC2. We then project the last 270 frames of each *60s* video onto this two-dimensional space. As shown in Fig. 1, a smaller distance between the generated and reference frames indicates less deviation from the pretrained distribution, while a smaller circle radius indicates lower feature variance and thus more stable generation. FreqForcing exhibits both the smallest distributional deviation and the lowest variance, demonstrating its superior stability and resistance to quality degradation during long-video generation.

Additional Ablation Studies

Ablation on Different Timestep Configurations. As discussed in the main paper, the global scene layout and object structure of the generated videos are primarily established during the early, high-noise denoising steps. We therefore apply SSA only at t_4 and t_3. To further validate this design choice, we evaluate SSA using different combinations of denoising steps, with the VBench-Long results reported in Tab. 1. Applying SSA to additional denoising steps yields no substantial performance improvement, while reducing generation throughput due to the increased computation of anchor attention.

Ablation on Different Anchor Cache Sizes. To investigate the effect of different anchor cache sizes on video generation, we conduct additional experiments with $N_{\text{anc}} \in \{3, 6, 9\}$ on VBench-Long. As shown in Tab. 2, $N_{\text{anc}} = 3$ achieves the best performance on visual consistency and image quality, whereas $N_{\text{anc}} = 9$ yields the highest dynamic degree and motion smoothness.

However, to provide a more intuitive evaluation of how the anchor cache size affects the generated frames, we further introduce two metrics: **Novelty** and **Repetitive Rate**. For video frame f_i, the definition of Novelty_i is as follows (Oquab et al. 2023):

$$F_i = \text{DINOv2}(f_i), \quad \text{Novelty}_i = 1 - \max_{0 \le j < i} \text{CosSim}(F_i, F_j). \tag{7}$$

Moreover, for a video containing N frames, the definition of **Repetitive Rate** is as follows:

$$\text{Repetitive\_Rate} = \frac{1}{N} \sum_{i=0}^{N-1} \left(\mathbf{1}_{\{\text{Novelty}_i < 0.1\}} \right). \tag{8}$$

Using these two metrics, we report the Novelty curves for different anchor cache sizes and Deep Forcing in Fig. 2 (a) and present the distributions of Repetitive Rate for videos generated by different methods in Fig. 2 (b). Although $N_{\text{anc}} = 3$ achieves the best performance on several VBench-Long metrics, it makes the model more prone to generating repetitive content over extended temporal horizons. Therefore, to balance quantitative performance and perceptual quality, we set $N_{\text{anc}} = 6$ as our final configuration.

Ablation on Different σ and λ Values for Spectral Anchoring. We have conducted additional experiments to investigate the impact of different σ and λ values for Spectral Anchoring in Tab. 2 of the main paper. To better illustrate the impact, we report the Novelty curves for different hyperparameter settings in Fig. 3 (a) and present the distributions of Repetitive Rate for videos generated by different methods in Fig. 3 (b). The results indicate that our proposed hyperparameter settings achieve a good balance between dynamic degree and visual consistency, while other settings may lead to either excessive repetition or reduced dynamic degree.

Quantitative Results on Causal Forcing. We report the VBench scores of applying FreqForcing to Causal Forcing (Zhu et al. 2026) in Tab 3. Our FreqForcing improves performance across all metrics, demonstrating its potential to generalize to other autoregressive video diffusion models.

More Visual Results

Additional qualitative comparisons on *120s* videos are presented in Fig. 4, Fig. 5, and Fig. 6. The results demonstrate that FreqForcing consistently outperforms representative autoregressive video generation methods across diverse scenarios, including dynamic scenes, complex backgrounds, and varying subject appearances. These visual results further validate the effectiveness of our proposed approach in maintaining video consistency and dynamic degree over extended temporal horizons.

Limitations and Future Work

Although FreqForcing effectively mitigates error accumulation in autoregressive video generation without additional training, several limitations remain. First, while spectral anchoring prevents the generated content from progressively drifting away from the desirable distribution during inference, some intermediate frames are still evicted from the KV cache. The loss of this historical context may affect subsequent generation, making more effective KV-cache management a promising direction for future research. Second, the dual-branch attention architecture introduces additional computation, resulting in computational overhead that grows linearly with the generation length and partially compromising the real-time capability of streaming video generation. Future work could investigate more computationally efficient mechanisms for maintaining long-horizon visual stability. Finally, owing to limited computational resources, we have not extended the core idea of FreqForcing to training-based approaches. Incorporating spectral anchoring into the training process represents another promising direction for future exploration.

Steps	Dynamic Degree↑	Motion Smoothness↑	Overall Consistency↑	Imaging Quality↑	Aesthetic Quality↑	Subject Consistency↑	Background Consistency↑
60s generation							
t_4	58.44	**98.31**	21.16	69.33	60.72	97.32	96.49
t_3, t_4 (Ours)	**58.87**	98.26	**21.17**	69.35	60.99	97.41	96.54
t_2, t_3, t_4	58.52	98.25	21.15	69.60	**61.15**	97.46	**96.58**
t_1, t_2, t_3, t_4	58.72	98.21	21.09	**69.92**	61.07	**97.53**	96.57

Table 1: **Quantitative comparison on different timestep configurations.** All metrics are reported on *60s* generations. The best results are highlighted in **bold**.

Anchor Cache Size	Dynamic Degree↑	Motion Smoothness↑	Overall Consistency↑	Imaging Quality↑	Aesthetic Quality↑	Subject Consistency↑	Background Consistency↑
60s generation							
$N_{\text{anc}} = 3$	57.59	98.19	**21.34**	**69.44**	**61.34**	**97.49**	**96.61**
$N_{\text{anc}} = 6$ (Ours)	58.87	98.26	21.17	69.35	60.99	97.41	96.54
$N_{\text{anc}} = 9$	**59.65**	**98.32**	21.09	69.41	60.80	97.39	96.52

Table 2: **Quantitative comparison on different anchor cache sizes.** All metrics are reported on *60s* generations. The best results are highlighted in **bold**.

Method	Dynamic Degree↑	Motion Smoothness↑	Overall Consistency↑	Imaging Quality↑	Aesthetic Quality↑	Subject Consistency↑	Background Consistency↑
60s generation							
Causal Forcing	63.01	96.45	16.90	62.54	51.39	95.11	95.49
Causal Forcing + FreqForcing	**87.10**	**96.54**	**20.07**	**68.16**	**58.91**	**95.89**	**95.58**

Table 3: **Quantitative comparison on Causal Forcing (Zhu et al. 2026) and our FreqForcing.** All metrics are reported on *60s* generations. The best results are highlighted in **bold**.

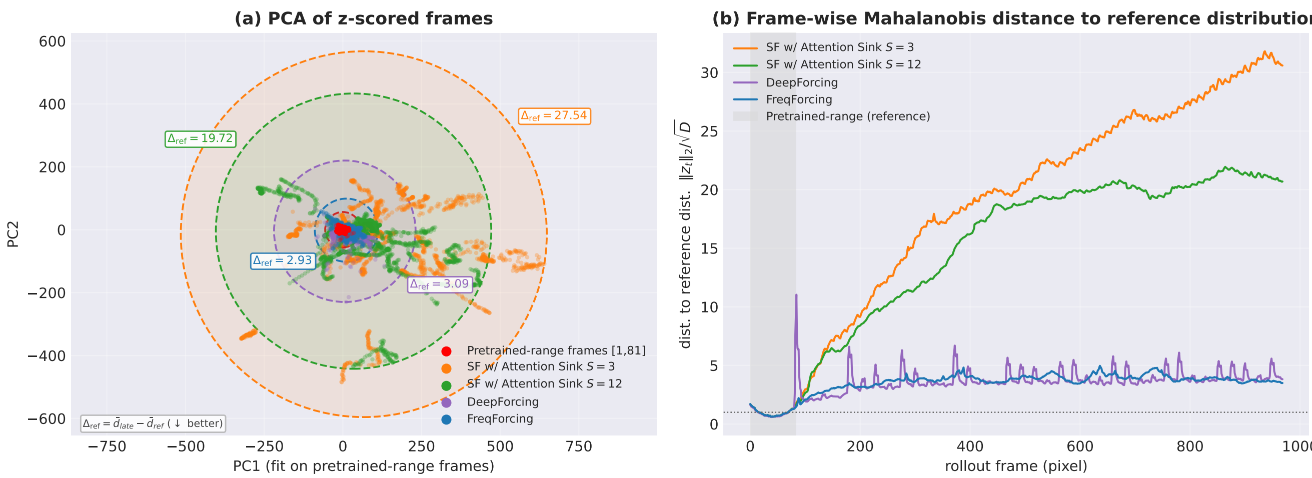

Figure 1: **Visualization of Distribution Shift.** **(a)** PCA visualization of different methods on *60s* videos. **(b)** Quantitative comparison of distribution shift, where the distance between the generated and reference frames is measured in the PCA space.

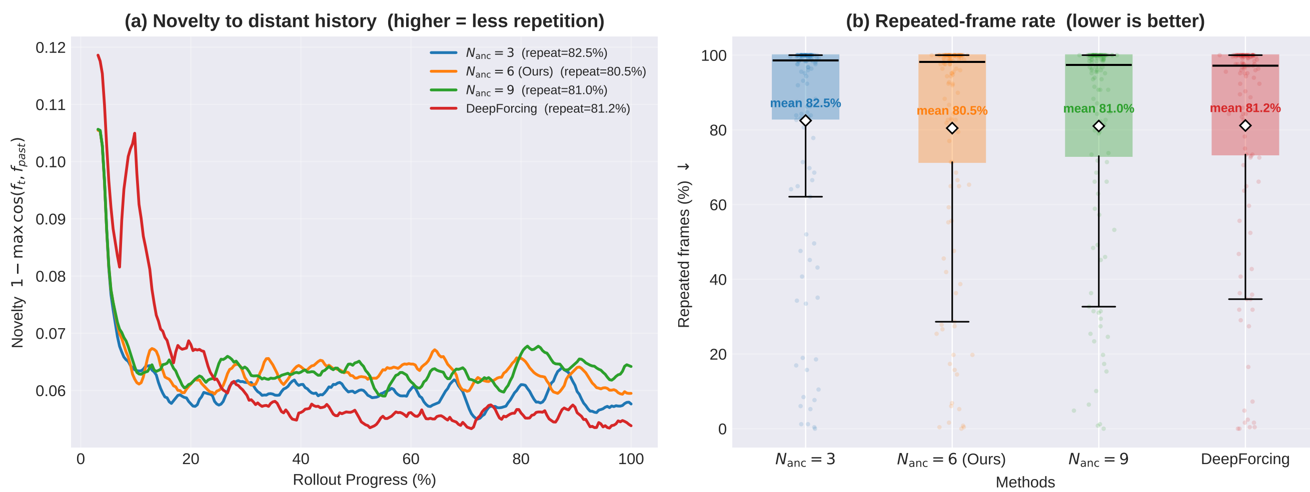

Figure 2: **Ablation study on different anchor cache sizes.** **(a)** We compare the novelty of the frames generated with different anchor cache sizes and Deep Forcing. **(b)** We report the Repetitive Rate for different anchor cache sizes and Deep Forcing.

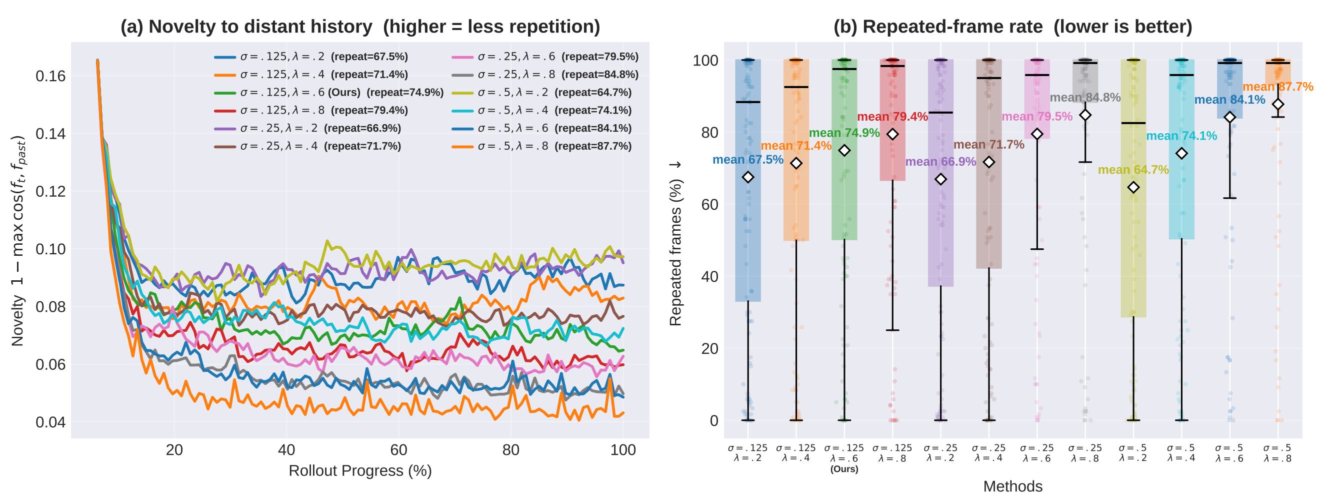

Figure 3: **Ablation study on different σ and λ values for Spectral Anchoring.** **(a)** We compare the novelty of the frames generated with different hyperparameter settings. **(b)** We report the Repetitive Rate for different hyperparameter settings.

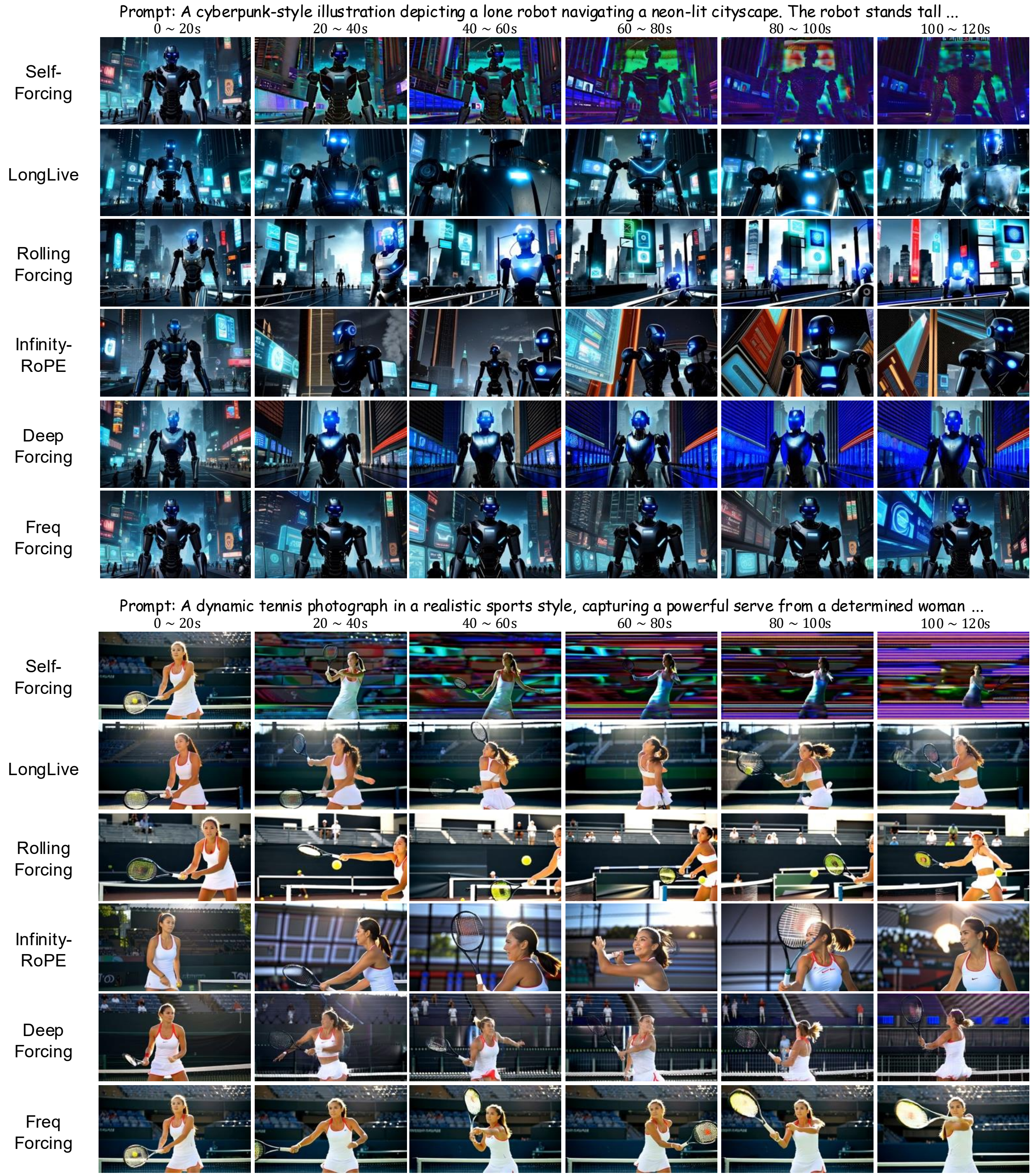

Figure 4: **Qualitative comparisons on *120s* videos.** We compare FreqForcing with representative autoregressive video generation methods on diverse scenarios.

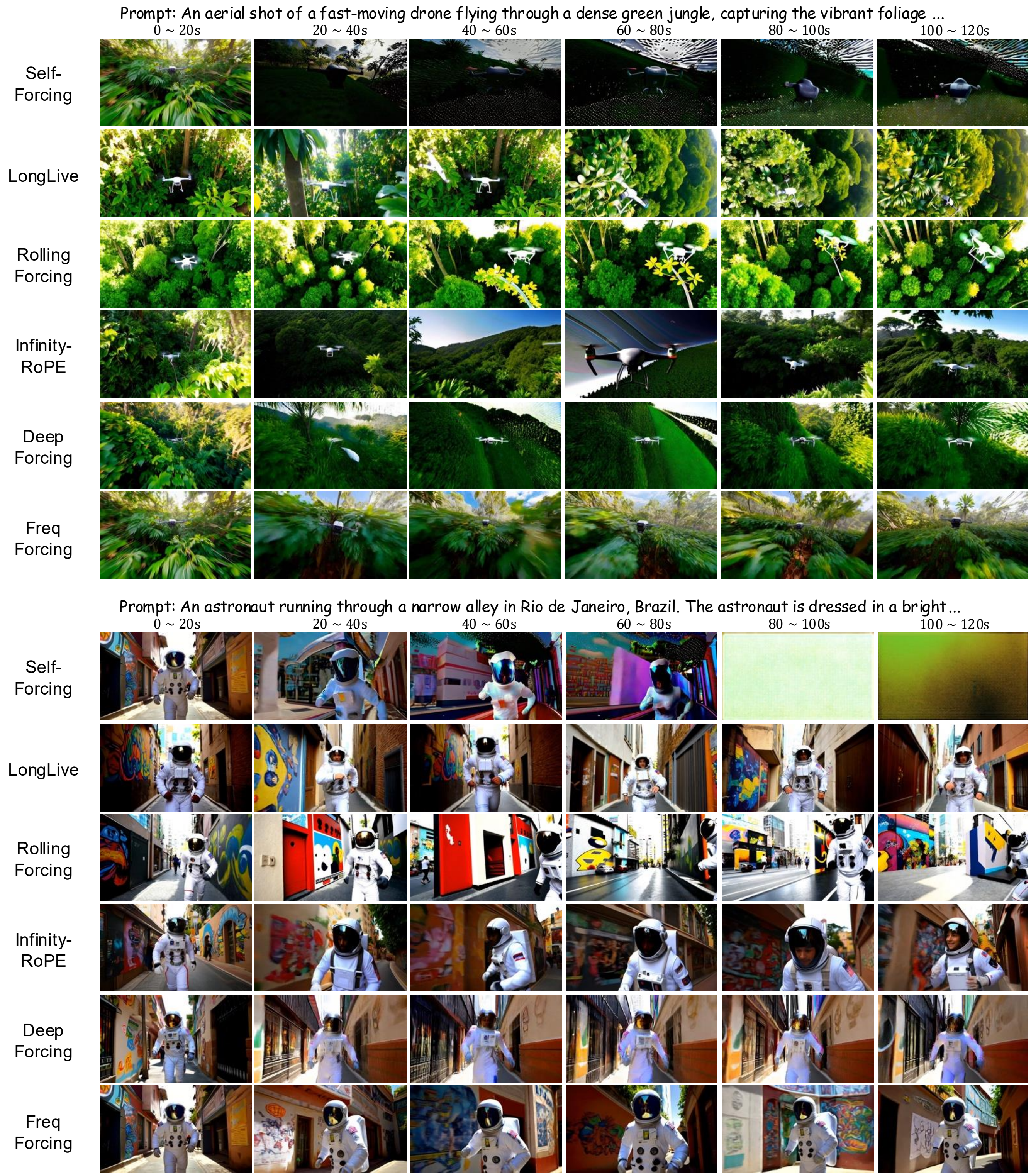

Figure 5: **Qualitative comparisons on *120s* videos.** We compare FreqForcing with representative autoregressive video generation methods on diverse scenarios.

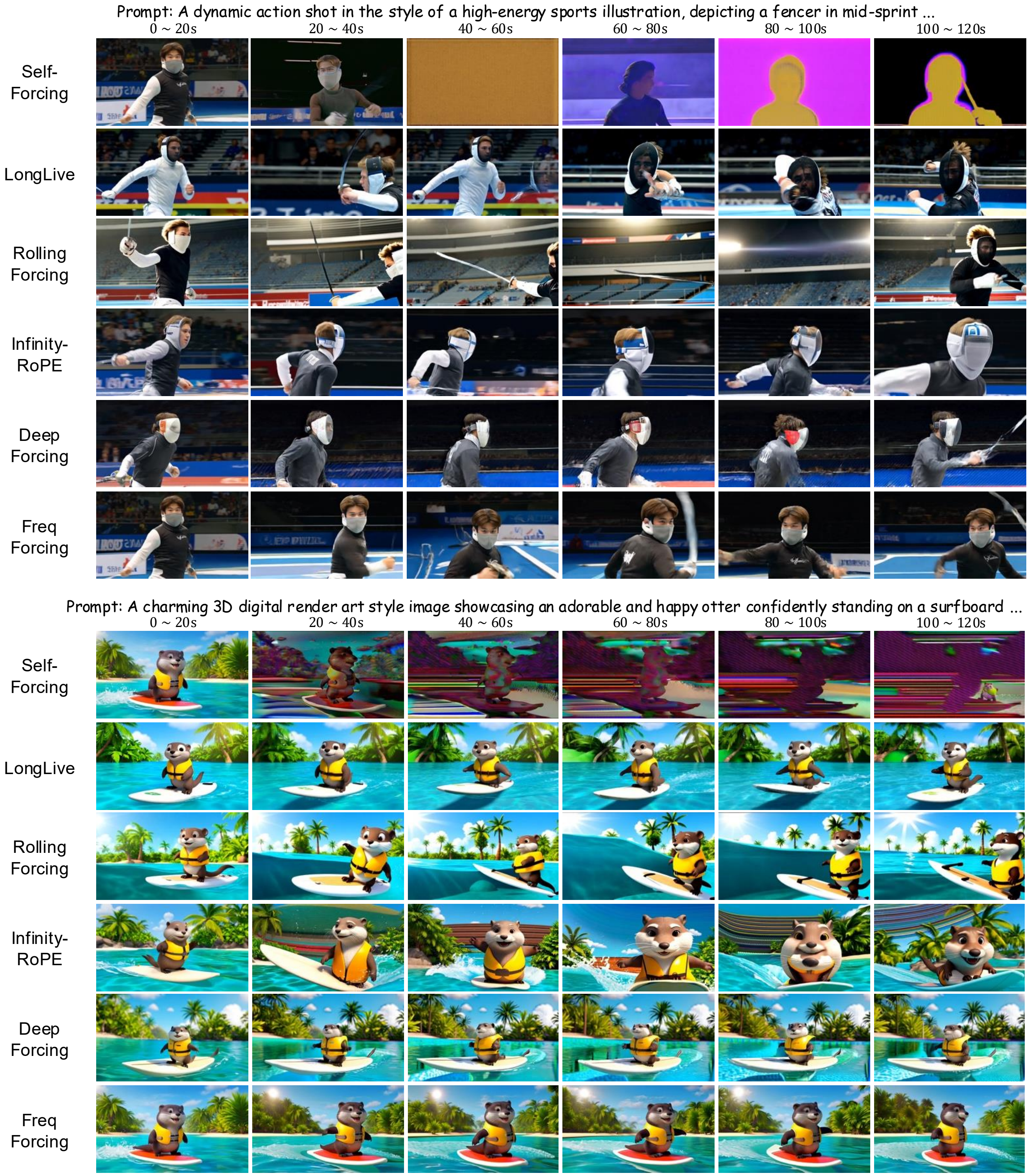

Figure 6: **Qualitative comparisons on *120s* videos.** We compare FreqForcing with representative autoregressive video generation methods on diverse scenarios.

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